Predicting Case Reoccurrence in Adult Protective Services

Dataset provided by APS, State of Michigan

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Our Predictive Pipeline: An Overview

This presentation will guide you through our methodology for tackling case reoccurrence.

Understanding the Challenge

Why this problem matters in Adult Protective Services.

Data Transformation

Structuring raw data and addressing missing values.

Establishing a Baseline

Performance with a Logistic Regression model.

Introducing XGBoost

Our novel, advanced machine learning approach.

Analyzing Performance

Comparing models and key insights gained.

Next Steps

Summary and future directions for this research.

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Why Does This Problem Matter?

The Domain

Adult Protective Services investigates abuse, neglect, and exploitation of vulnerable adults — but case closure doesn't mean the danger is over.

The Challenge

Reoccurrence means prior interventions failed. Agencies lack tools to identify which closed cases carry the highest risk of returning.

The Goal

Build a classification model to predict reoccurrence, enabling proactive resource allocation to the most at-risk clients.

Dataset Overview

It’s a dataset from Adult Protective Services (APS), State of Michigan. The data is a de-identified list of all APS clients of any age who were referred to APS between Jan 1, 2022 and Dec 31, 2024. This includes:

- referrals not opened for investigation (Disposition Status Desc = denied)
- referrals assigned for investigation (Disposition Status Desc = open, closed, soft closed)
- include any case open in the 3 year time frame that may have had an open/referred case prior to Jan 1, 2022 (Prior_APS_History_Before_2022 columns is a review of history between 2019-01-01 and 2021-12-31)
- each client was given a unique ID number
- Null data is represented by “?” within the entire Excel workbook.

The dataset originally contained 191148 rows x 154 columns.

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First look on the raw data

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	T	U	V	W	X	Y
ID	Referral_Date	Open_Date	Closed_Date	Number_c	Disposition Status	DENY_RE	CLOSURI	Prior_AP	AGE_at_ti	GENDER	RACE	Marital St	Veteran Ind	Housing_	Does_	d_Harm_vs_	Alleged_Ha	Investiga	Alleged_H	Investigat	Alleged_H	Investiga	Alleged_
987679398-1	29.03.2024	?	?	?	DENIED	Client has	?	Y	80	M	Caucasiar	?	N	Rental	N		N	?	N	?	N	?	N
987679398-2	04.09.2024	04.09.2024	19.09.2024	15	CLOSED	?	Services N	Y	80	M	Caucasiar	Single-Nev	N	Rental	N		N	Not Applic	Y	Unsubstan	N	Not Applic	N
987679482-1	01.03.2022	01.03.2022	15.04.2022	45	CLOSED	?	Situation s	N	57	M	Caucasiar	Married-Liv	N	Own Hom	N		Y	Unsubstai	Y	Unsubstan	N	Not Applic	N
987679482-2	04.03.2022	?	?	?	DENIED	Duplicate r	?	N	57	M	Caucasiar	Married-Liv	N	Own Hom	N		N	?	Y	?	N	?	N
987679482-3	13.02.2023	13.02.2023	13.04.2023	59	CLOSED	?	Situation s	N	58	M	Caucasiar	Single-Nev	N	Own Hom	N		Y	Unsubstai	N	Not Applica	N	Not Applic	N
987679482-4	12.07.2023	?	?	?	DENIED	Client has	?	N	58	M	Caucasiar	Married-Liv	N	Own Hom	N		N	?	N	?	N	?	N
987679482-5	30.11.2023	?	?	?	DENIED	Client has	?	N	58	M	Caucasiar	Single-Nev	N	Own Hom	N		Y	?	N	?	N	?	N
987679482-5	30.11.2023	?	?	?	DENIED	Client has	?	N	58	M	Caucasiar	Single-Nev	N	Own Hom	N		Y	?	N	?	N	?	N
987679482-6	19.12.2023	?	?	?	DENIED	Client has	?	N	58	M	Caucasiar	Married-Liv	N	Own Hom	N		N	?	N	?	N	?	N
987679482-7	09.10.2024	09.10.2024	07.11.2024	29	CLOSED	?	Situation s	N	?	M	Caucasiar	Married-Liv	N	Own Hom	N		N	Not Applic	Y	Unsubstan	N	Not Applic	N
987679482-8	20.11.2024	?	?	?	DENIED	Client has	?	N	59	M	Caucasiar	Single-Nev	N	Own Hom	N		N	?	Y	?	N	?	N
987679482-9	09.12.2024	?	?	?	DENIED	Client has	?	N	59	M	Caucasiar	Single-Nev	N	Own Hom	N		Y	?	N	?	N	?	N
987679491-1	07.05.2024	?	?	?	DENIED	Has been	?	N	60	M	Caucasiar	Single-Nev	N	Own Hom	N		N	?	N	?	N	?	N
987679491-2	08.05.2024	?	?	?	DENIED	Has been	?	N	60	M	Caucasiar	Single-Nev	N	Rental	Y		N	?	N	?	N	?	N
987679504-1	03.02.2022	?	?	?	DENIED	Has been	?	Y	61	F	Caucasiar	Single-Nev	N	AFC/HA	?		N	?	N	?	N	?	N
987679506-1	31.10.2024	31.10.2024	20.12.2024	50	CLOSED	?	Services r	N	56	F	Caucasiar	Married-Liv	N	Own Hom	N		N	Not Applic	Y	Substantial	N	Not Applic	N
987679507-1	05.04.2024	?	?	?	DENIED	Has been	?	N	62	M	Caucasiar	Single-Nev	N	Rental	Y		N	?	N	?	N	?	N
987679549-1	27.07.2023	?	?	?	DENIED	Client is n	?	N	55	M	Caucasiar	Married-Liv	N	Own Hom	N		N	?	N	?	Y	?	N
987679558-1	08.06.2023	08.06.2023	21.06.2023	13	CLOSED	?	Situation s	N	69	M	Caucasiar	?	N	Own Hom	Y		Y	Unsubstai	Y	Unsubstan	N	Not Applic	N
987679616-1	05.04.2024	?	?	?	DENIED	Client has	?	N	64	F	Caucasiar	Single-Nev	N	Own Hom	Y		N	?	N	?	N	?	N
987679616-2	29.08.2024	?	?	?	DENIED	Client is n	?	N	64	F	Caucasiar	Divorced	N	Own Hom	Y		Y	?	N	?	N	?	N
987679646-1	14.12.2022	14.12.2022	23.01.2023	40	CLOSED	?	Situation s	Y	58	F	Caucasiar	Single-Nev	N	Home of A	N		N	Not Applic	N	Not Applica	N	Not Applic	N
987679646-2	31.03.2023	31.03.2023	27.04.2023	27	CLOSED	?	Situation s	Y	58	F	Unable to	Single-Nev	N	Home of A	N		N	Not Applic	Y	Unsubstan	N	Not Applic	N
987679670-1	11.01.2022	11.01.2022	10.02.2022	30	CLOSED	?	Situation s	Y	81	F	Caucasiar	Widowed	N	Own Hom	N		Y	Unsubstai	Y	Unsubstan	N	Not Applic	N
987679670-2	16.02.2022	16.02.2022	14.03.2022	26	CLOSED	?	Situation s	Y	81	F	Caucasiar	Widowed	N	Home of A	N		N	Not Applic	Y	Unsubstan	N	Not Applic	N
987679725-1	16.08.2024	?	?	?	DENIED	Has been	?	N	60	M	Caucasiar	Single-Nev	N	Other	Y		N	?	Y	?	N	?	N
987679756-1	17.10.2022	17.10.2022	26.10.2022	9	CLOSED	?	Situation s	N	64	M	Caucasiar	Divorced	N	Rental	Y		N	Not Applic	Y	Unsubstan	N	Not Applic	N
987679781-1	28.11.2022	?	?	?	DENIED	Client has	?	N	45	M	Caucasiar	?	N	Own Hom	N		Y	?	Y	?	N	?	N
987679781-1	28.11.2022	?	?	?	DENIED	Client has	?	N	45	M	Caucasiar	?	N	Own Hom	N		Y	?	Y	?	N	?	N
987679818-1	15.08.2022	?	?	?	DENIED	Client has	?	N	39	M	Caucasiar	?	N	Own Hom	N		N	?	N	?	N	?	N
987679831-1	12.12.2024	?	?	?	DENIED	Has been	?	Y	54	M	Caucasiar	Single-Nev	N	AFC/HA	N		N	?	N	?	N	?	N
987679831-2	13.12.2024	?	?	?	DENIED	Has been	?	Y	54	M	Caucasiar	Single-Nev	N	AFC/HA	N		N	?	N	?	N	?	N
987679839-1	21.07.2022	21.07.2022	21.10.2022	92	CLOSED	?	Refused S	N	44	M	Caucasiar	Single-Nev	N	Own Hom	Y		Y	Substantia	N	Substantial	N	Not Applic	N
987679839-1	21.07.2022	21.07.2022	21.10.2022	92	CLOSED	?	Refused S	N	44	M	Caucasiar	Single-Nev	N	Own Hom	Y		Y	Substantia	N	Substantial	N	Not Applic	N
987679839-2	19.11.2024	19.11.2024	17.12.2024	28	CLOSED	?	Client Ref	N	46	M	Caucasiar	Single-Nev	N	Own Hom	Y		Y	Substantia	N	Not Applica	N	Not Applic	N

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Structuring the Chaos

```python
# Removing the Refferal Only Cases
df = data[data['open_date'].notna()]
print(f'original shape: {data.shape}')
print(f'new shape: {df.shape}')
```

✓ 1.3s

```
original shape: (191148, 154)
new shape: (78559, 154)
```

```python
print(f"Original columns: {data.shape[1]}")
print(f"Safe Working columns: {df_model.shape[1]}")
```

✓ 0.0s

```
Original columns: 154
Safe Working columns: 97
```

From Raw Records to Actionable Data

The raw dataset contained **191,000+ unstructured records**. After rigorous filtering, **78,559 active investigations** remained — each representing a real client and a real risk.

Leakage Prevention

Stripped all aggregate timeline columns and post-investigation "close" metrics the model could not know at *t_0*.

```python
# 5 & 6. Create stop_time and event_next
# Get the next case's open_date
df['next_open_date'] = df.groupby('person_id')['open_date'].shift(-1)

# event_next: 1 if another APS case occurs, 0 otherwise
df['event_next'] = df['next_open_date'].notna().astype(int)

# Define end of observation period as 3 years after t0
df['end_of_obs'] = df['t0'] + pd.DateOffset(years=3)

# Calculate stop_time (in days)
df['stop_time'] = np.where(
    df['event_next'] == 1,
    (df['next_open_date'] - df['t0']).dt.days,
    (df['end_of_obs'] - df['t0']).dt.days
)
```

Survival Analysis variable

Restructured chronologically using `stop_time` and `event_next` to properly evaluate case-level risk over time.

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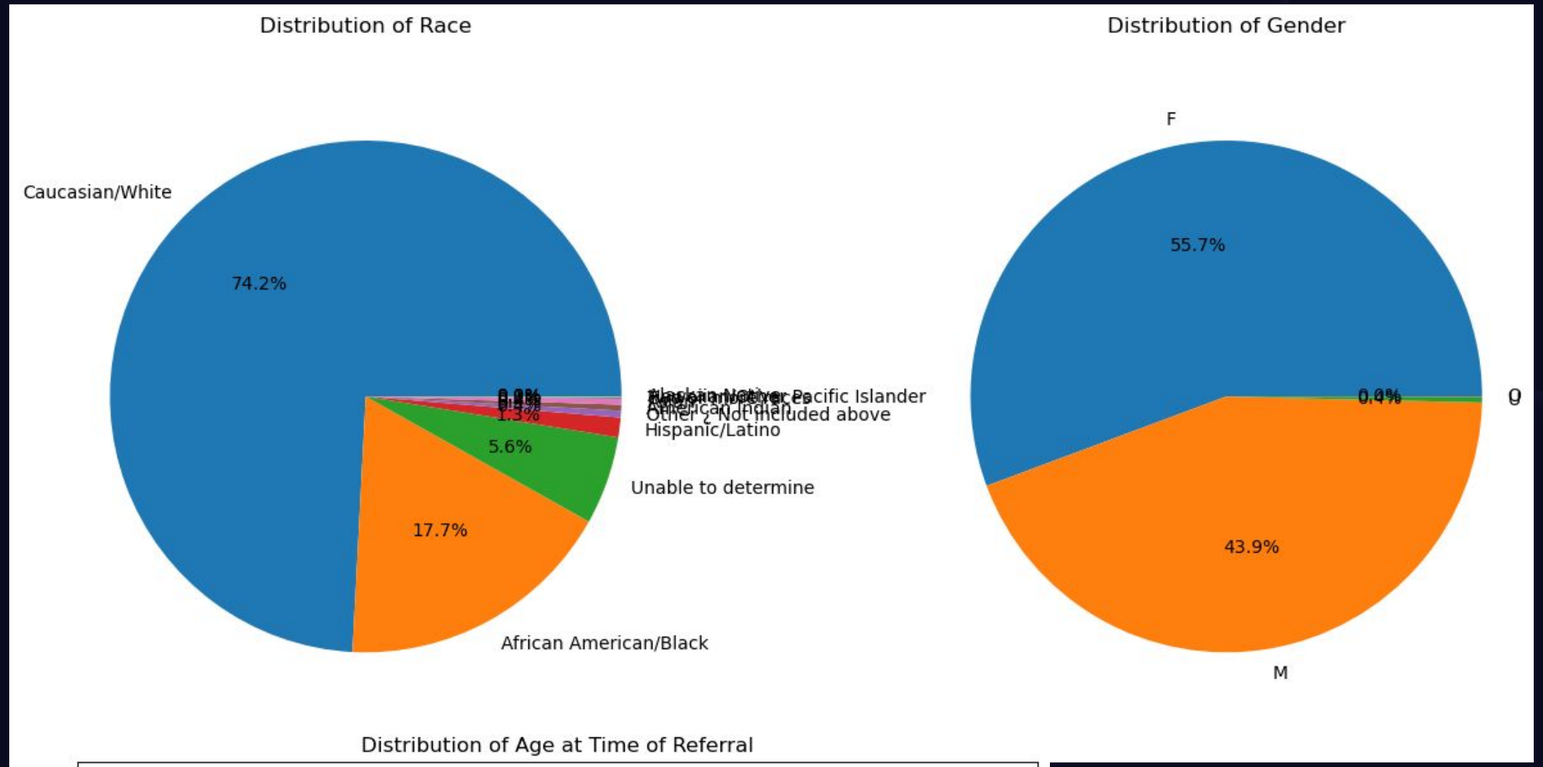

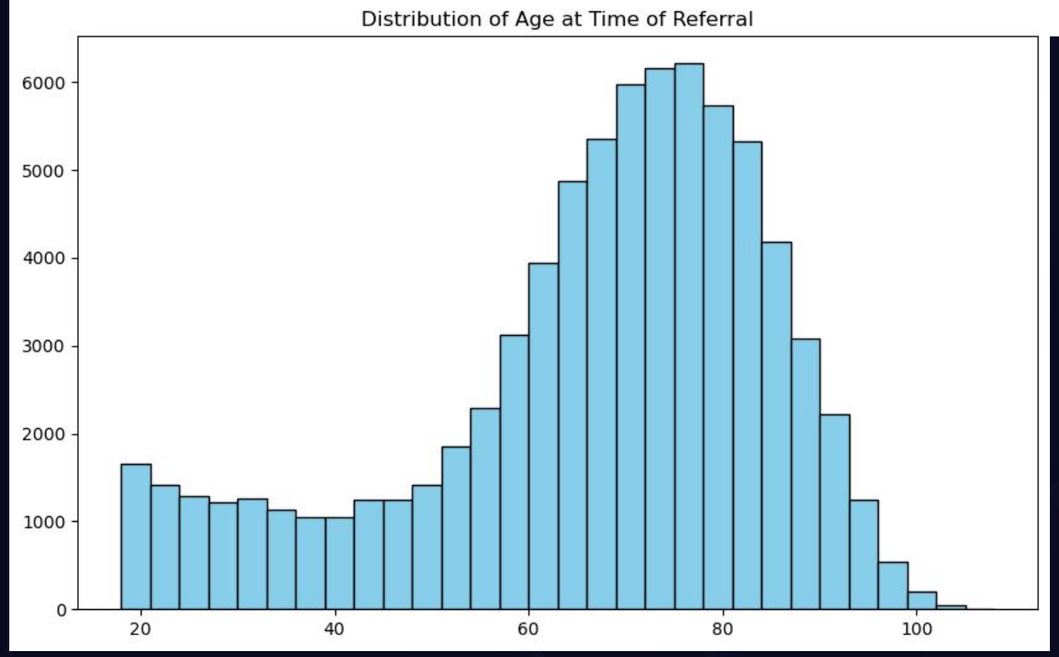

Categories:

Alleged columns: 14
Investigation columns: 8
Verified columns: 6
Client risk columns: 14
Environmental columns: 9
Support network columns: 6
Caregiver columns: 10
Perpetrator columns: 4
Demographic columns: 8
Case Info columns: 14
Reoccurrence columns: 4

Overall: 97

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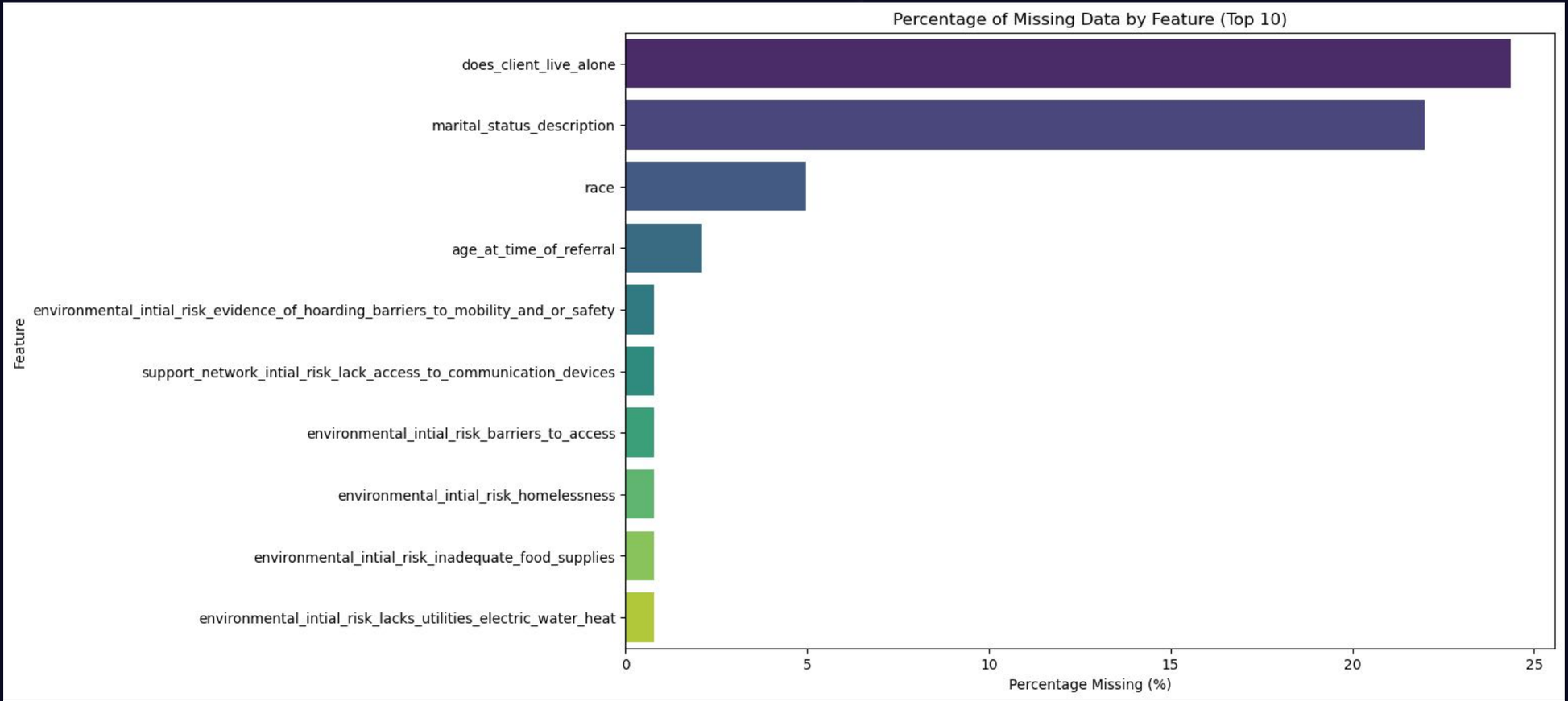

	Feature	Missing_Count	%_Missing
12	does_client_live_alone	19138	24.36
9	marital_status_description	17271	21.98
8	race	3899	4.96
6	age_at_time_of_referral	1650	2.10
59	environmental_intial_risk_evidence_of_hoarding...	610	0.78
68	support_network_intial_risk_lack_access_to_com...	610	0.78
58	environmental_intial_risk_barriers_to_access	610	0.78

The proportion of Race roughly matches the demographic data in Michigan according to the Demographic data in https://www.micalhoun.org/demographicdata?id=25&sectionId=940

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Weaponizing the Missing Data

Standard Advice: Drop or Impute

In APS data, **missingness is itself a signal**. (does_client_live_alone)

⚠ **MNAR Reality:** Missing marital status correlated with severe Self-Neglect — caseworkers prioritized acute emergencies over paperwork.

The Engineering Solution

Structural voids were encoded as an explicit **Unknown class**, teaching the model that *"missing paperwork = elevated danger."*

→ Identify MNAR patterns across demographic fields

→ Engineer structural voids into learnable categories

→ Let algorithms treat absence as information

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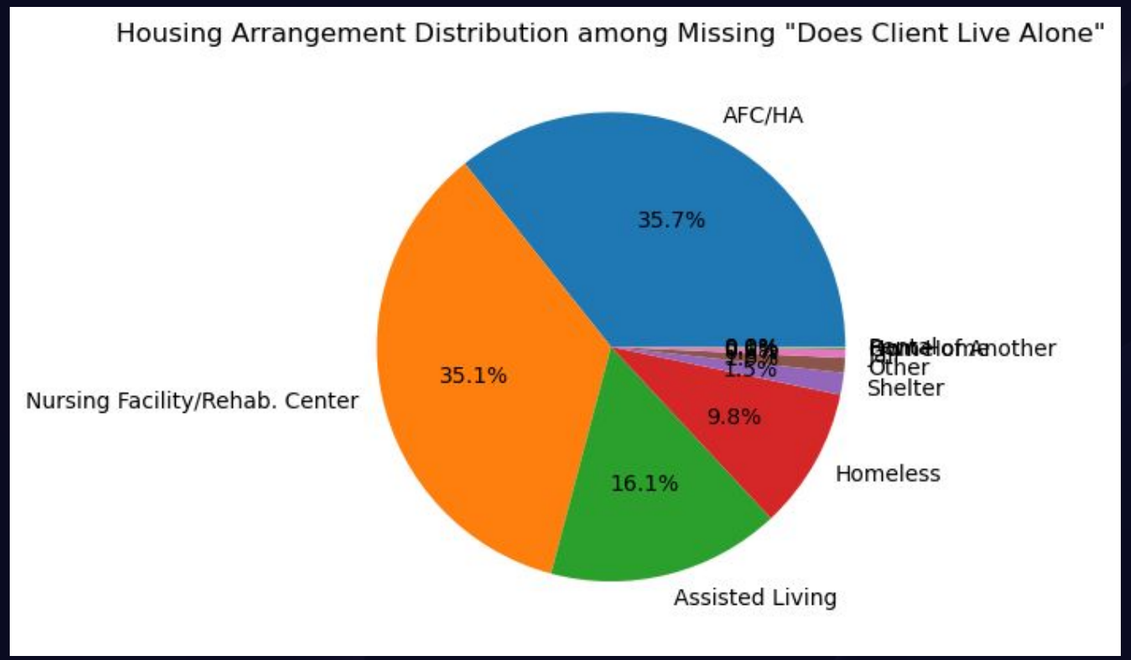

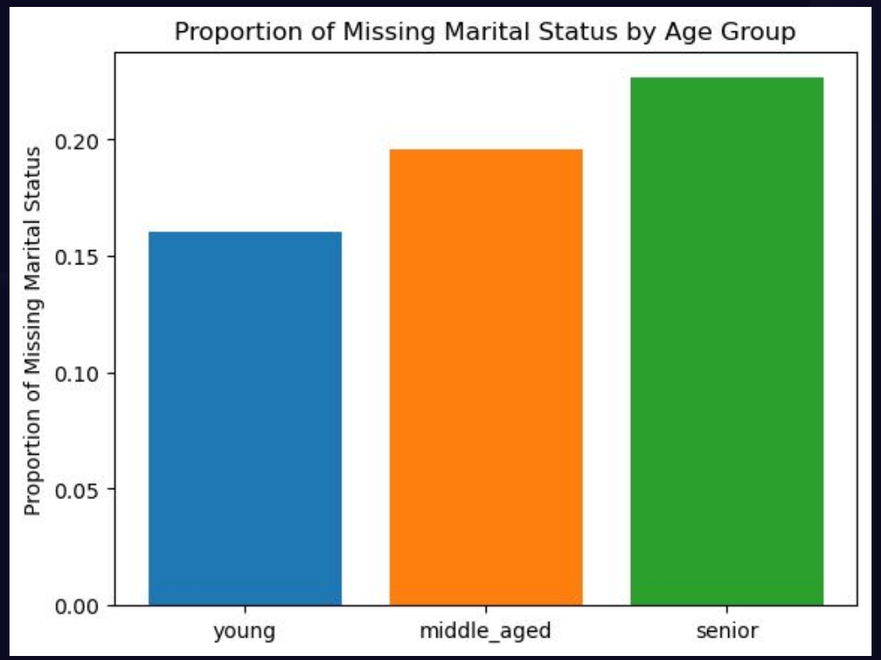

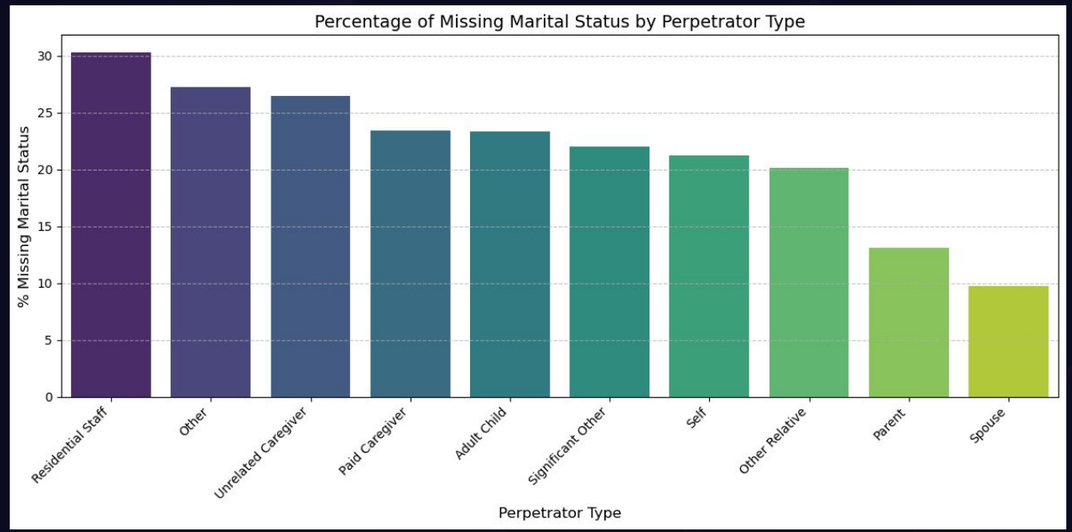

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Feature Engineering

#1

id -> person_id & case_order
open_date -> t0 & next_open_date

#2

Engineered stop_time (observation duration) and reoccurred (*binary target, later renamed to event_next*) to evaluate chronological risk and prevent data leakage.

#3

Preserved MNAR demographic data by engineering Unknown class features (e.g., marital_status_Unknown) to mathematically capture "Crisis Triage" chaos.

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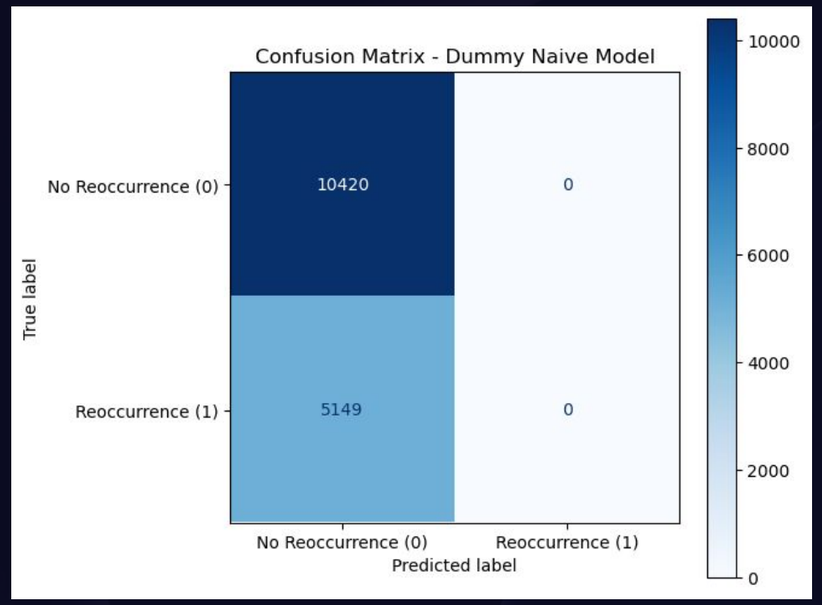
The Baseline: Dummy (naive model)



Confusion Matrix

What the Numbers Reveal

A naïve model guessing "No Reoccurrence" every time achieves ~67% accuracy but catches **zero** at-risk cases.

```
Accuracy: 0.6693

Classification Report:
              precision    recall  f1-score   support

           0       0.67      1.00      0.80     10420
           1       0.00      0.00      0.00      5149

    accuracy                           0.67     15569
   macro avg       0.33      0.50      0.40     15569
weighted avg       0.45      0.67      0.54     15569
```

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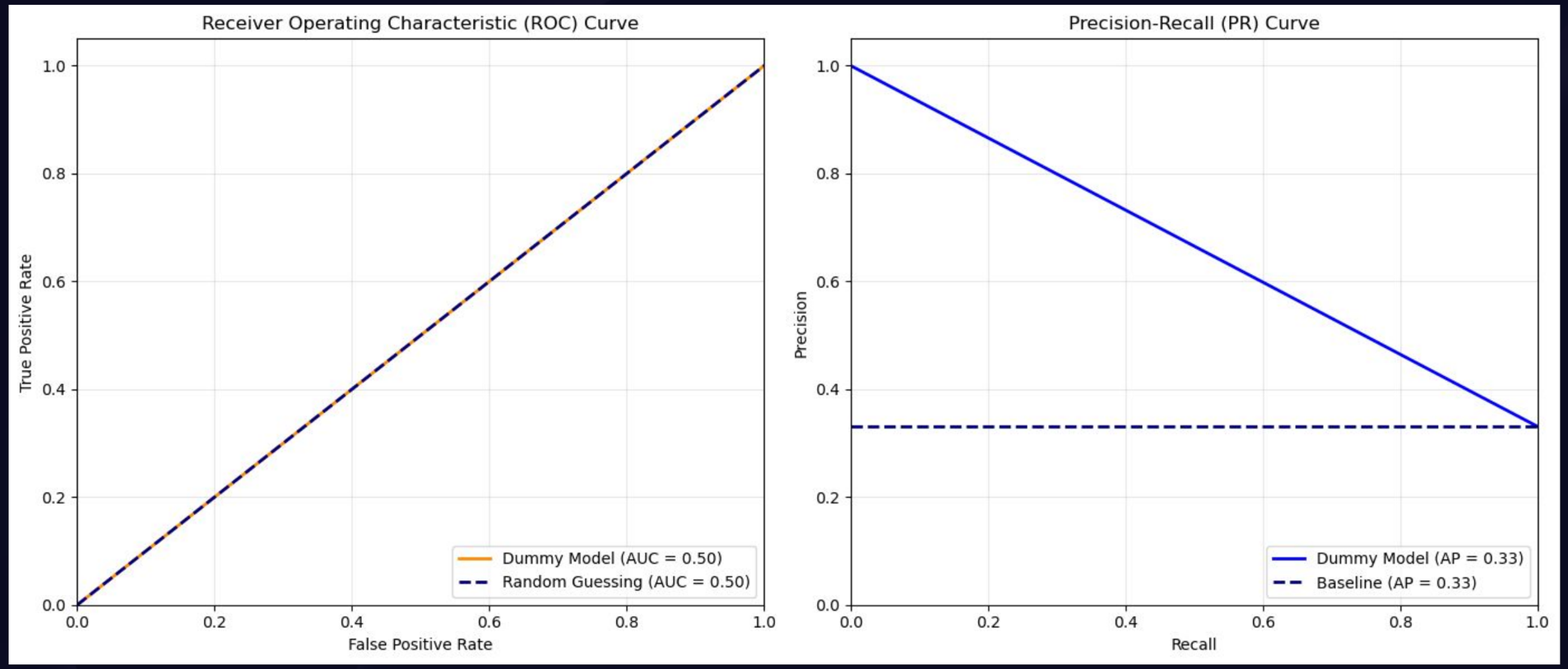
Naive Model



ROC Curve & Precision_Recall Curve

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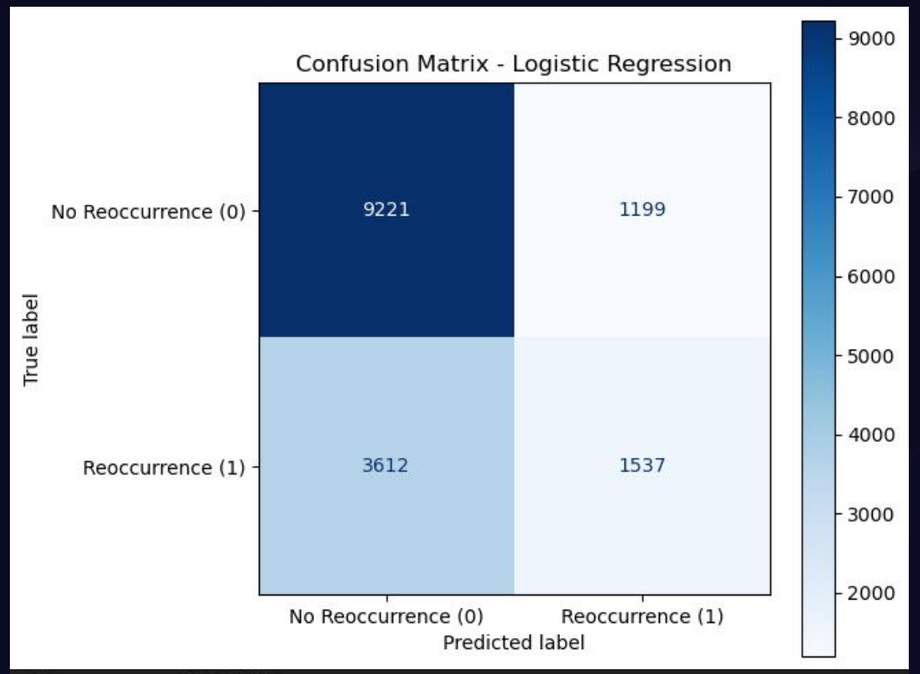


```
Accuracy: 0.6910

Classification Report:
              precision    recall  f1-score   support

           0       0.72      0.88      0.79     10420
           1       0.56      0.30      0.39      5149

    accuracy                           0.69     15569
   macro avg       0.64      0.59      0.59     15569
weighted avg       0.67      0.69      0.66     15569

ROC-AUC Score: 0.7005
Average Precision (PR-AUC): 0.5113
```

Logistic Regression

What the Numbers Reveal

Logistic Regression improved to **69% accuracy** and **ROC-AUC of 0.70** — but as a linear model, it struggled with class imbalance and sparse one-hot encoded data.

Critical Gap: Class 1 Recall of only **0.30** means 3,612 vulnerable adults were missed — a failure the next model must address.

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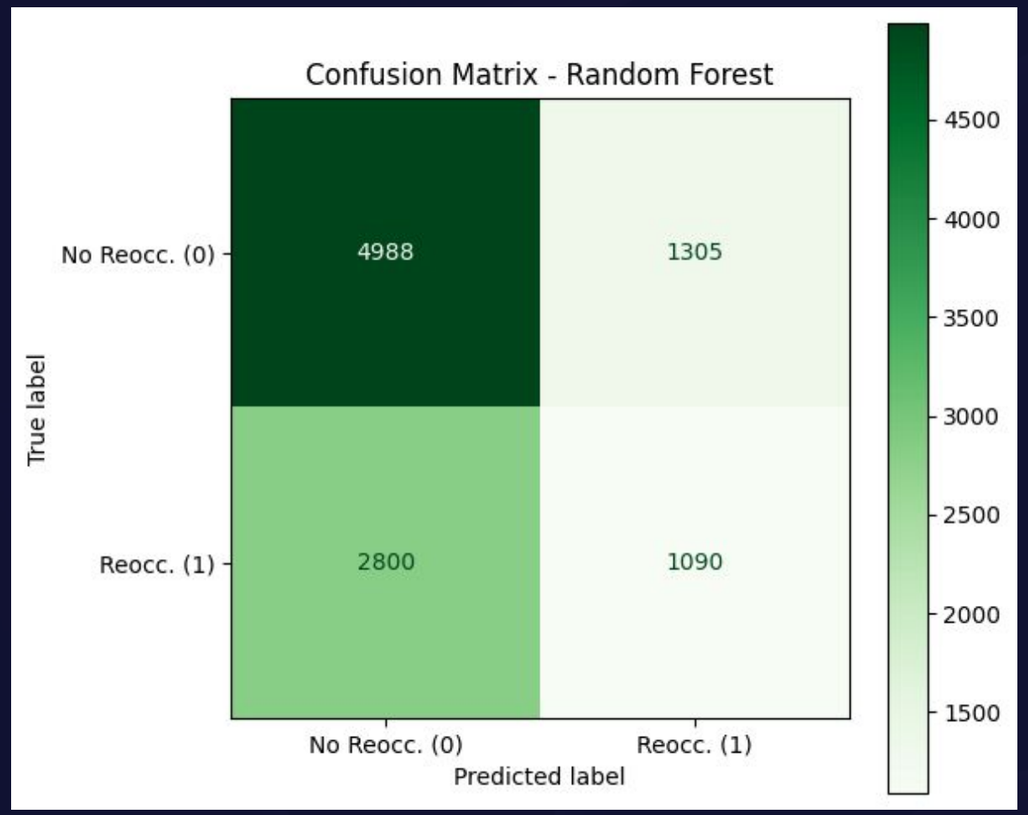
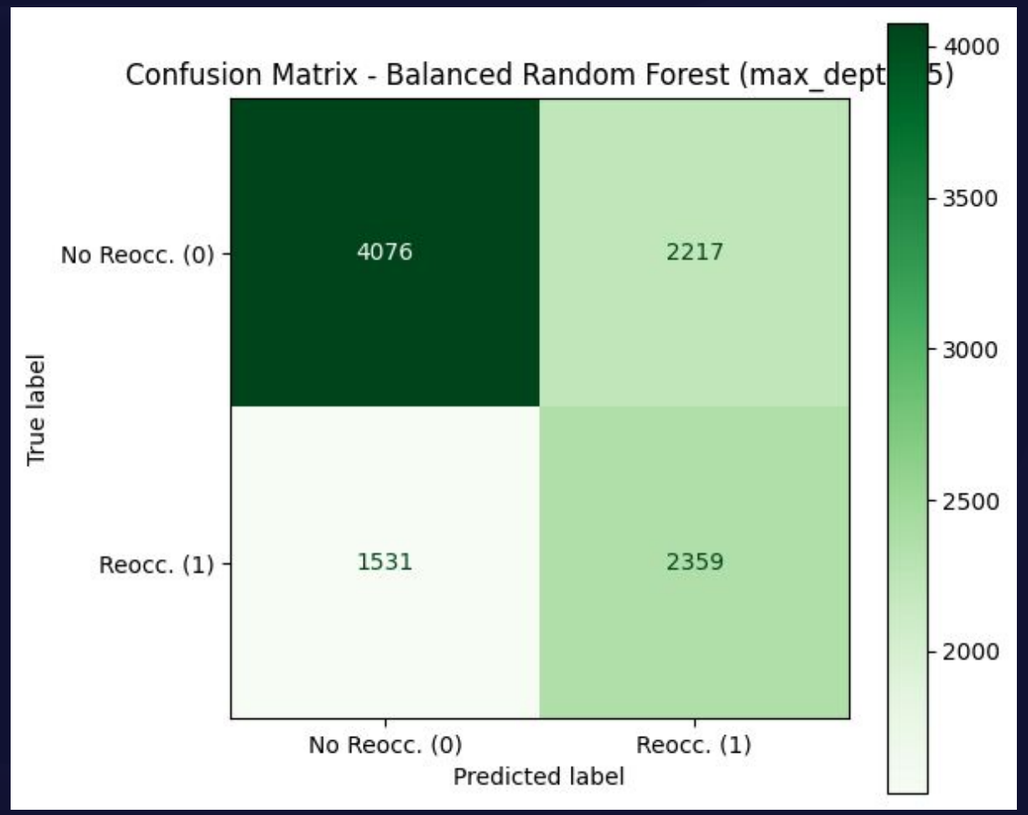
Random Forest Performance

Standard Model (The Lazy Tree)



Baseline analysis showing significant majority-class bias.

Balanced Model (Max Depth 5)



Improved minority class capture through parameter balancing.
Increased the penalty by setting the class weight = balanced.

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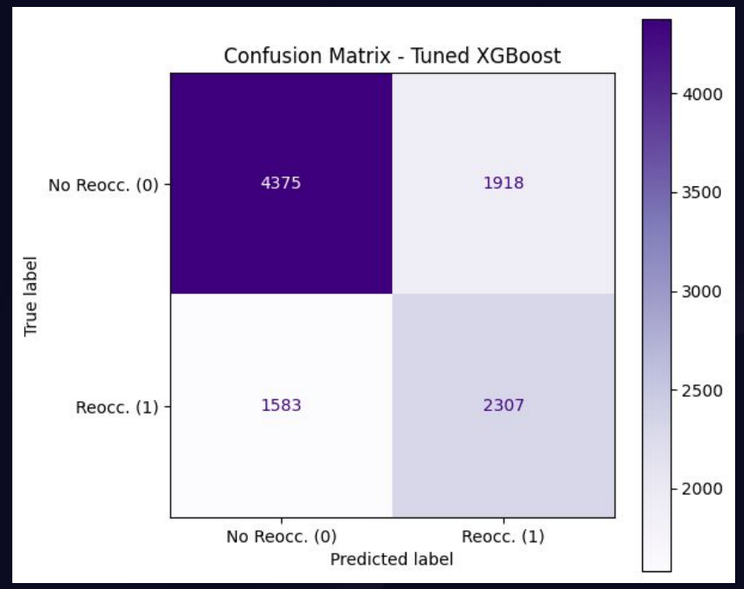
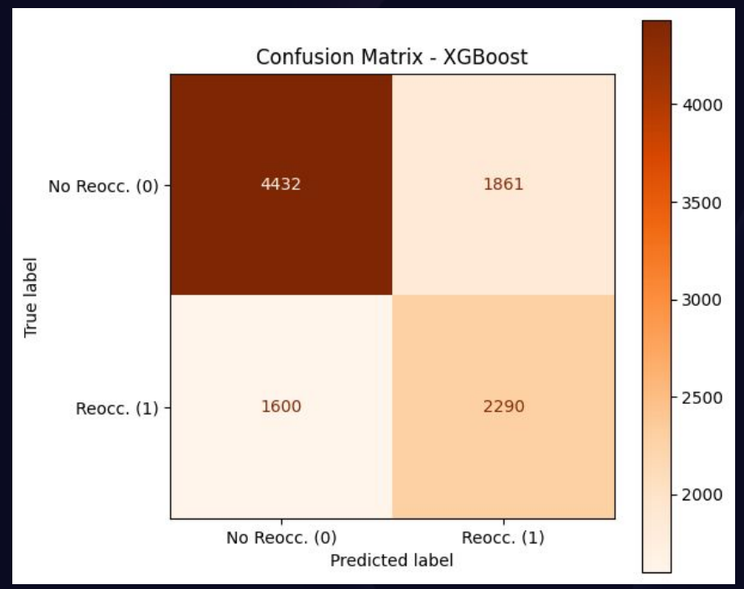
XGBoost Performance





Accuracy: 66%

Gradient boosting mathematically evaluates "Gain" rather than "Gini impurity," allowing it to handle your hundreds of sparse, One-Hot Encoded columns without getting distracted by continuous variables like age.

```
Classification Report:
              precision    recall  f1-score   support

           0       0.73      0.70      0.72      6293
           1       0.55      0.59      0.57      3890

    accuracy                           0.66     10183
   macro avg       0.64      0.65      0.64     10183
weighted avg       0.66      0.66      0.66     10183

ROC-AUC Score: 0.6885
Average Precision (PR-AUC): 0.5491
```

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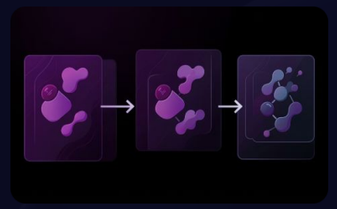
The Novel Model: XGBoost

Why Random Forest Failed

Class 1 Recall dropped to **0.24**. Majority-class bias and Gini impurity inflated continuous variables like age, drowning out the minority signal.



The XGBoost Upgrade

Extreme Gradient Boosting mathematically handles wide, sparse matrices. Sequential error correction targets the weaknesses of each prior learner.

The Strategy

Heavily penalized **False Negatives** to force the algorithm to prioritize the minority class — catching real danger over optimizing overall accuracy.

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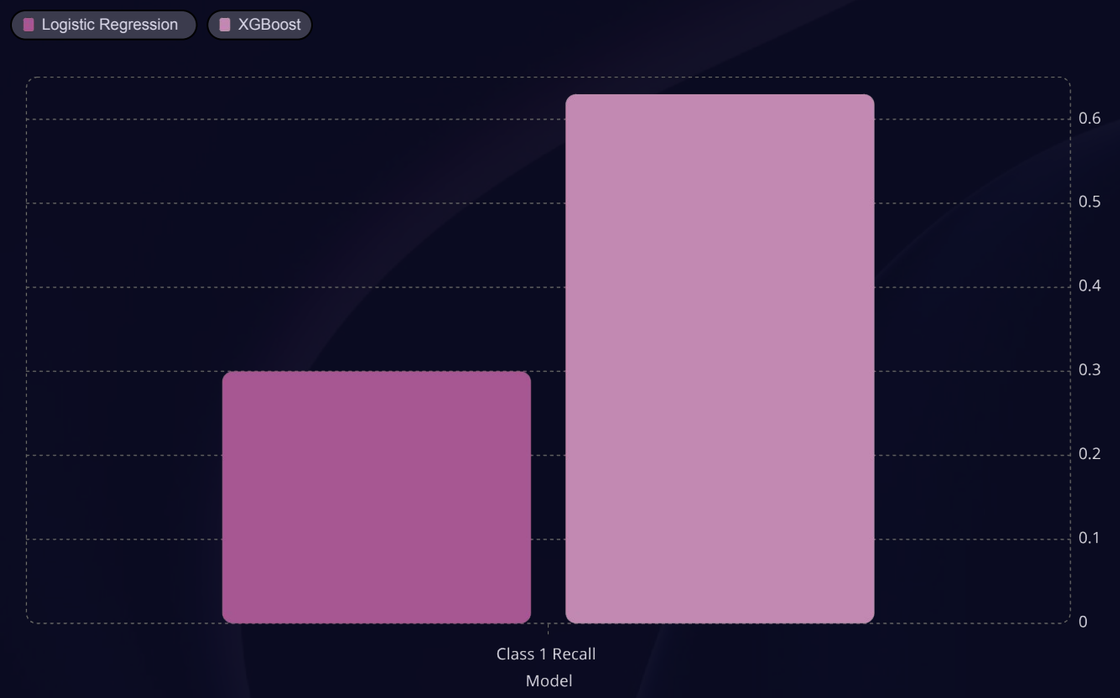
Results: Shattering the Baseline



The Trade-off That Paid Off

A marginal dip in overall accuracy was accepted to **prioritize the minority class** — the clients who actually return to APS.

This is not an abstract metric. Each additional case caught represents a real person who may receive help before a crisis escalates.

0.63 Recall

Class 1 Recall more than doubled from the logistic baseline.

3,254 Caught

Returning vulnerable adults identified — up from 1,810 under the baseline.

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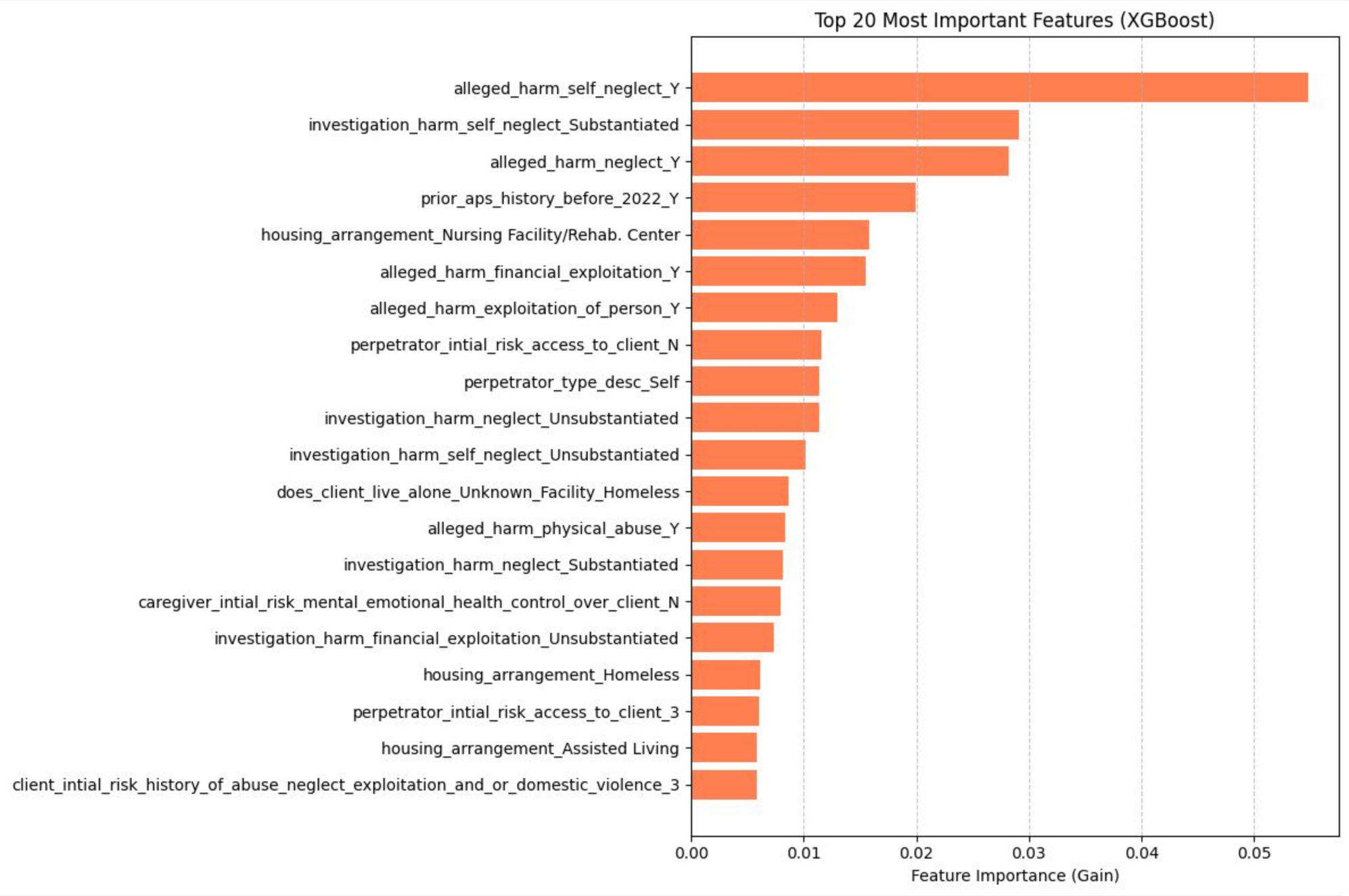

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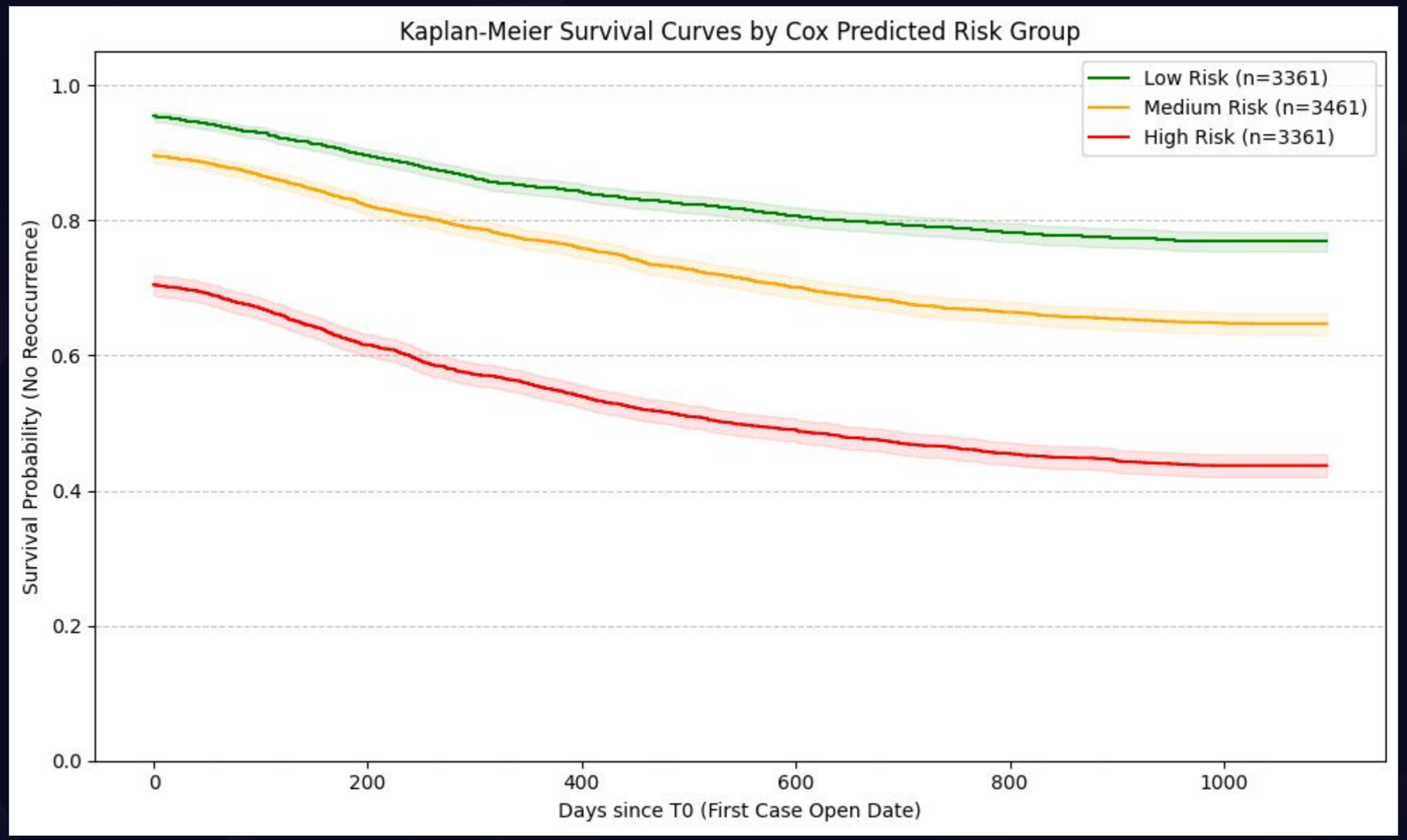
Cox model: Survival Analysis



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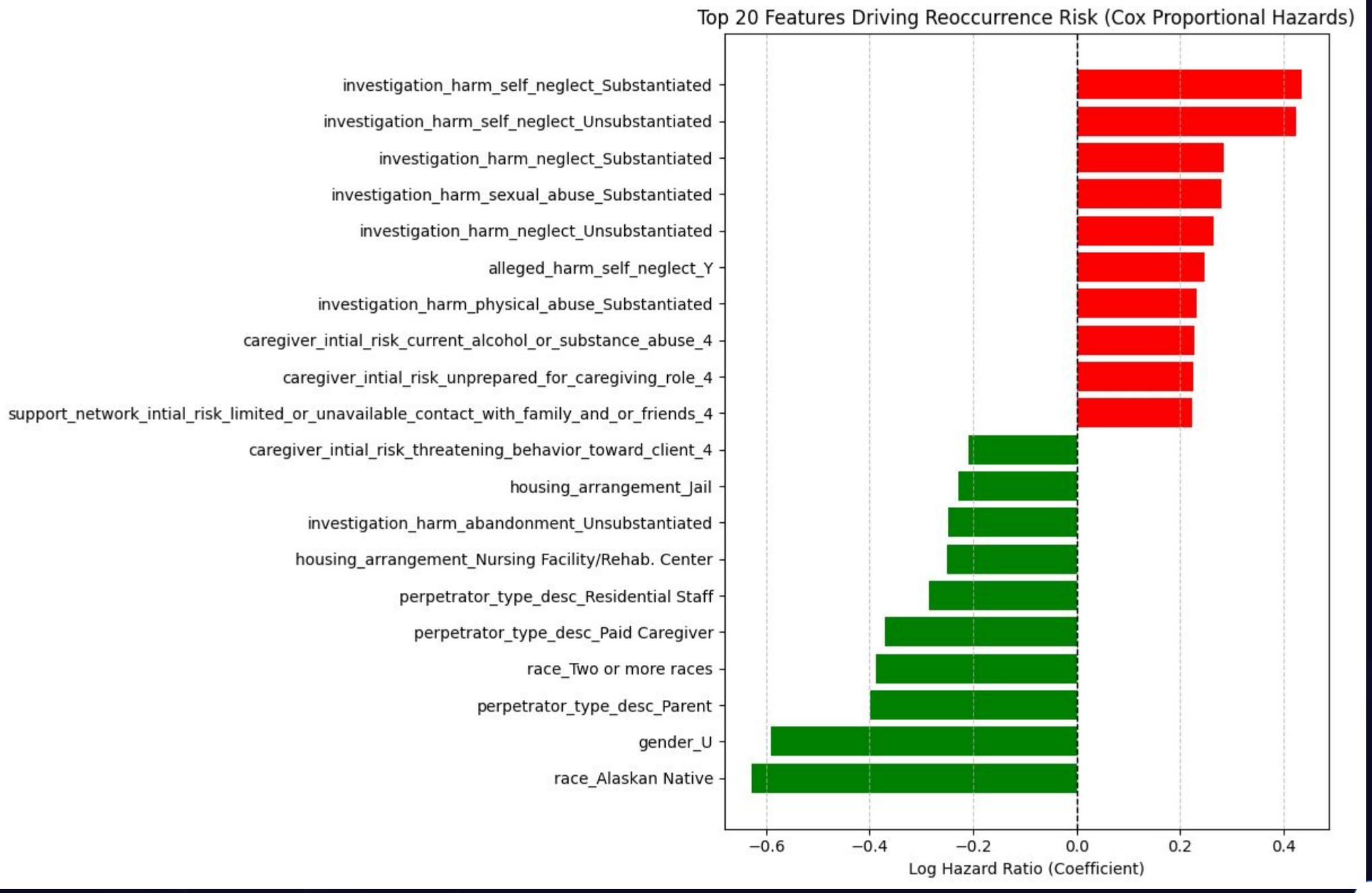

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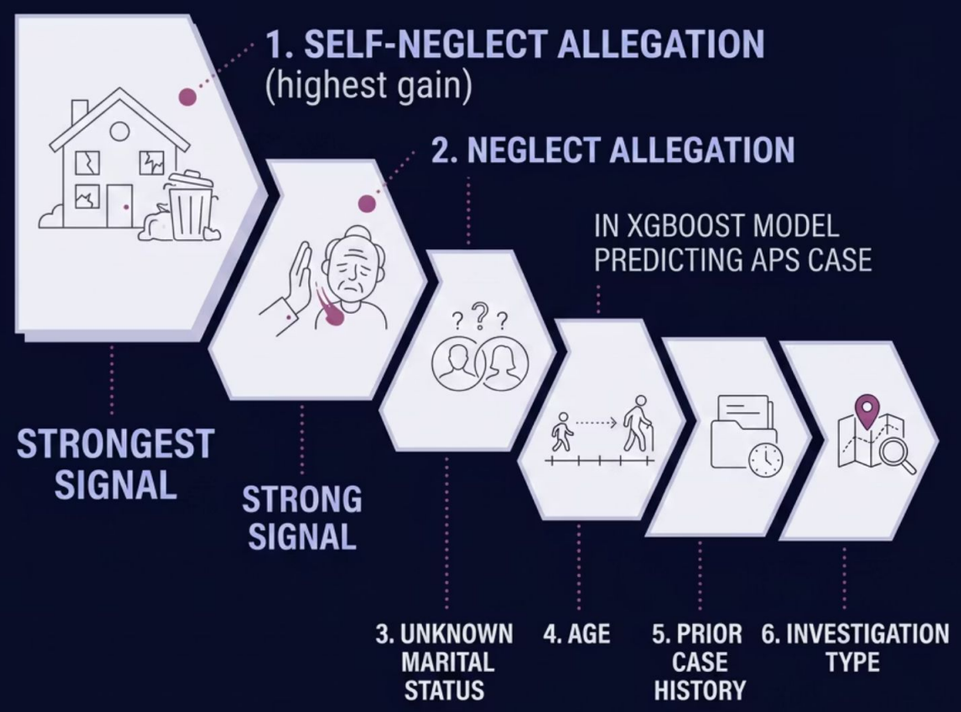
Key Insights: "Smoke Means Fire"



Self-Neglect and Neglect dominate as the strongest signals of future reoccurrence.

The Domain Finding

The top predictors of future reoccurrence were initial allegations of **Self-Neglect and Neglect** — even when those cases were ultimately marked *"Unsubstantiated."*

> **Interpretation:** The mere fact that an investigator had to open a self-neglect case signals an unstable environment. Legal thresholds for substantiation don't erase real-world risk.

This finding validates the core thesis: **administrative chaos contains predictive signal** — if you know how to read it.

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Conclusion & Next Steps

1

What We Proved

Machine learning can interpret administrative chaos to protect vulnerable populations — without dropping data or ignoring leakage.

2

Immediate Impact

XGBoost more than doubled Class 1 Recall, identifying 3,254 returning vulnerable adults that a linear baseline would have missed.

3

Future: Cox Proportional Hazards

Transition from binary classification to continuous **time-to-event modeling**, evaluating Hazard Ratios to properly resolve right-censored data.